

Merged

One chain, two decisions

*The combined physics, parameters and objective for the Baghewala digital twin
— source for the deck, the simulator, and model training.*

PROBLEM STATEMENT

SIH26120 · Oil India Limited · Software · Smart Automation

USE FOR

Presentation content · Simulator · ML parameter ranges · Judge Q&A

OPEN ITEMS

Three, listed in section six

Revised 12 September 2026 after a document review. Oil SG and the Walther constant A are now stated exactly (equation 3). The float-limit timing is corrected (equation 5). Open item 1 is rewritten: the viscosity curves agree at reservoir temperature and spread most at steam temperature, not the other way round. Missing constants are added (equation 1). New tier-D assumptions: the cooling-constant law (equation 2), the production-rate law and pump displacement (equation 4), the injection and soak phases (equation 7) and the measured signals (equation 8). The cumulative-oil definition is corrected. Steam ranges are narrowed to Baghewala field data. Reservoir pressure is flagged as uncertain. The pump-intake equation is corrected. Symbol clashes are removed. An unsupported error figure is removed.

One

The problem

Baghewala, Rajasthan. Heavy crude at 14–19° API, roughly 11,500 cP at 50 °C (the midpoint of the reported 10,000–13,000 cP), in a reservoir with a native temperature of 46–50 °C. At that thickness it will not flow, and reservoir pressure is too low to push it anyway. Two interventions are mandatory: steam to thin the oil, and a sucker rod pump to lift it.

Steam heats the near-wellbore rock. Over the following weeks that heat drains into the surrounding cold formation, viscosity climbs back, and two things degrade at once — less oil reaches the well, and the rod string can no longer fall through the thickening oil on its downstroke. It goes slack, gets shock-loaded on every stroke, and fatigues until it fractures.

Both effects have one cause: reservoir temperature. Steam scheduling and pump operation are currently managed by separate teams on historical practice, so nothing connects them.

That disconnect is the problem statement. The deliverable is a well-to-surface digital twin that observes the current state, predicts how it evolves, and prescribes both decisions from the same prediction.

Two

The causal chain

Steam volume + quality → heated zone V_h → temperature $T(t)$ → viscosity μ

then splits

↓ mobility → productivity J_o → inflow q_{in} → oil rate

↑ rod damping → maximum safe pump speed N_{max}

both converge on

SOR — barrels of steam per barrel of oil

The split is the key structural point. Production potential and mechanical safety are not two problems that happen to share a well. They are two consequences of one variable, which is why optimising them separately is guaranteed to be suboptimal.

Heated zone volume matters beyond its obvious role: a larger heated zone has proportionally less surface exposed to cold rock, so it cools more slowly. Equation 2 makes this exact — the cooling constant follows the zone's volume-to-surface ratio. Steam volume therefore buys *duration*, not just temperature. This is the physical variable through which "inject more now" versus "wait" becomes computable rather than habitual.

Three

The objective

Maximise oil produced per unit of steam injected, subject to pump speed staying below the rod-floating limit.

Stated formally, over a cycle:

OBJECTIVE FUNCTION

$$\min \text{ SOR} = V_{\text{steam}} / N_p \quad \text{subject to} \quad N(t) \leq N_{\text{max}}(\mu(t))$$

Decision variables: steam volume V_s , soak time t_{soak} , re-steam date t_{cut} , and pump speed $N(t)$ each day.

N_p = cumulative oil over the cycle, $N_p = \int \min(q_{\text{in}}, Q_{\text{pump}}) dt$, where Q_{pump} is the pump displacement at the set speed (equation 4). Do not multiply q_{in} by fillage: low fillage is the result of low inflow, so that counts the same limit twice. SOR uses total N_p , which includes the oil the well would give cold.

Two consequences worth stating explicitly in the deck.

Rod safety is a constraint, not the goal. If avoiding failures were the objective, the optimum would be to run at minimum speed forever — no failures, almost no oil. Failures matter because they halt production and incur workover cost. The value of knowing N_{max} precisely is that it lets the pump run *close* to the limit rather than conservatively far below it.

Re-steam timing moves SOR more than pump speed does. Pump speed is a daily optimisation within a cycle. Re-steam timing is the cycle-level decision, and it directly sets the numerator of SOR. Too early wastes steam on a well still producing; too late means weeks of crawling output at 4 SPM.

Four Equations

1 · STEAM → HEATED ZONE [ASSUMPTION — REDUCED-ORDER ENERGY BALANCE]

$$V_h = \eta \cdot m_s \cdot [x \cdot L + c_p(T_s - T_i)] / [M_R(T_s - T_i)]$$

Steam enthalpy divided by rock heat requirement gives heated rock volume. Radius $r_h = \sqrt{(V_h / \pi h)}$.

Constants: $m_s = \rho_w \cdot V_s$ with $\rho_w = 1,000 \text{ kg/m}^3$, because V_s is a cold-water-equivalent volume. $L = 1,716.6 \text{ kJ/kg}$, the latent heat of steam at 250 °C. $c_p = 4.187 \text{ kJ/(kg} \cdot \text{K)}$. $M_R = 2,350 \text{ kJ/(m}^3 \cdot \text{K)}$. $T_i = 50 \text{ °C}$. $h = 9.92 \text{ m}$ net heated thickness (tier D: back-solved from the worked radius; replace with field net pay).

$T_s = 250 \text{ °C}$ is an effective downhole value (tier D). Field surface steam is 280–305 °C; the downhole temperature after wellbore losses is not reported. L is held at its 250 °C value.

Worked: 2,000 m³ CWE, $x = 0.70$, $T_s = 250 \text{ °C}$, $T_i = 50 \text{ °C}$ gives 2,039 kJ/kg. With $\eta = 0.5$: $V_h = 4,338 \text{ m}^3$, $r_h = 11.8 \text{ m}$.

Feeds: equation 2, through the heated radius r_h that sets the cooling constant τ ; and equation 4, through r_h in the two-zone rate law.

2 · THERMAL BALANCE → TEMPERATURE [PHYSICS; COOLING LAW IS AN ASSUMPTION]

$$C_{\text{eff}} \cdot dT_r/dt = \dot{Q}_{\text{steam}} - \dot{Q}_{\text{loss}} - \dot{Q}_{\text{prod}}$$

General form, valid during injection, soak and production. For the production phase with steam off and loss proportional to the temperature gap (heat carried out by the produced fluid is folded into the loss term), this integrates to a closed form usable directly in a dashboard:

$$T(t) = T_i + (T_s - T_i) \cdot e^{-t/\tau} \quad \text{where } \tau = C_{\text{eff}}/U_{\text{eff}}$$

Cooling-constant law [assumption, tier D]: τ is proportional to the heated zone's volume-to-surface ratio. For a cylinder of radius r_h and height h that ratio is $r_h \cdot h / 2(r_h + h)$, so

$$\tau = \tau_{\infty} \cdot r_h / (r_h + h), \quad \tau_{\infty} = 30 \cdot (r_{h,\text{ref}} + h) / r_{h,\text{ref}} = 55.2 \text{ days}$$

$r_{h,\text{ref}} = 11.80 \text{ m}$ is the worked radius, so $\tau = 30$ days exactly in the worked case. τ rises with steam volume and levels off towards 55 days; there is no hard cap. Over the simulator range (500–4,000 m³) τ runs from 21 to 35 days.

Worked ($\tau = 30 \text{ d}$; t is the production day, and day 0 is the first day after soak): day 0 → 250 °C · day 30 → 124 °C · day 60 → 77 °C · day 90 → 60 °C.

3 · TEMPERATURE → VISCOSITY [PHYSICS — ASTM D341 / WALTHER, ONE CONSTANT CALIBRATED]

$$\log_{10}(\log_{10}(v + 0.7)) = A - B \cdot \log_{10}(T)$$

T in kelvin, v in cSt, μ (cP) = v × SG with **SG = 0.963** (from $\rho_o = 963 \text{ kg/m}^3$; about 15.4 °API, inside the field range). Anchored on 11,500 cP at 50 °C — the midpoint of the reported 10,000–13,000 cP, not a single measurement — with B = 3.5.

A is computed from the anchor and not rounded: $A = \log_{10}(\log_{10}(11,500/0.963 + 0.7)) + 3.5 \cdot \log_{10}(323.15) = 9.3933$. Rounding A to 9.394 gives 11,684 cP at the anchor, 1.6% high.

Worked: day 0 → 5 cP · day 30 → 93 cP · day 60 → 1,148 cP · day 90 → 4,465 cP · fully cold (50 °C) → 11,500 cP

Do not substitute a simple exponential. Oil viscosity changes too steeply with temperature for one exponential to hold from 50 °C to 250 °C. The double-logarithm form (ASTM D341) is the standard for petroleum liquids.

4 · VISCOSITY → PRODUCTION [PHYSICS; RATE LAW AND PUMP CONSTANTS ARE ASSUMPTIONS]

$$J_o \propto k \cdot k_{ro} / \mu_o \quad \cdot \quad q_{in} = J_o (P_r - P_{wf})$$

Thin oil flows more readily, so the productivity index rises as viscosity falls. Inflow is productivity times drawdown.

Rate law [assumption, tier D — Boberg–Lantz two-zone radial flow]: the heated zone (r_w to r_h , viscosity $\mu_h(t)$) and the cold zone (r_h to r_e , viscosity μ_c) resist flow in series. Drawdown is held constant in v1.

$$q_{in}(t) = q_{cold} \cdot \ln(r_e/r_w) / [(\mu_h(t)/\mu_c) \cdot \ln(r_h/r_w) + \ln(r_e/r_h)]$$

$r_w = 0.1 \text{ m}$, $r_e = 100 \text{ m}$, $q_{cold} = 10 \text{ bbl/d}$ (the unstimulated rate per well), $\mu_c = \mu(T_i) = 11,500 \text{ cP}$, $\mu_h(t)$ from equation 3 at T(t). When the heated oil is thin, the cold zone still limits flow, so the rate has a natural ceiling of $q_{cold} \cdot \ln(r_e/r_w) / \ln(r_e/r_h) = 3.23 \times q_{cold}$ at $r_h = 11.8 \text{ m}$. As the zone cools the rate falls back to q_{cold} , not to zero. q_{cold} is set so the worked case lands inside the reported SOR band.

Pump displacement [assumption, tier D]: $Q_{pump} = A_p \cdot S \cdot N \cdot 1440 \cdot \eta_v$ in m^3/d (N in strokes per minute, 1,440 minutes per day). Plunger 1.75" ($A_p = 1.552 \times 10^{-3} \text{ m}^2$), volumetric efficiency $\eta_v = 0.8$, pump setting depth $D_p = 1,100 \text{ m}$. v1 uses the polished-rod stroke S for the plunger stroke.

Surface rate: $q = \min(q_{in}, Q_{pump})$ at the set speed N. Slowing the pump costs oil only when Q_{pump} falls below q_{in} .

Worked ($2,000 \text{ m}^3$, $x = 0.70$, $S = 2.5 \text{ m}$, $N = 6 \text{ SPM}$): $Q_{pump} = 168.6 \text{ bbl/d}$, so the reservoir limits the rate all cycle. $q = 32.3 \text{ bbl/d}$ on day 0 · 31.7 on day 30 · 26.4 on day 60 · 17.3 on day 90 · 12.7 on day 120. Over 120 production days (trapezoidal rule, 1-day steps) $N_p = 2,946 \text{ bbl}$. Steam = $2,000/0.159 = 12,579 \text{ bbl}$, so **SOR = 4.27**.

Coupled to the pump by: pump intake $P_{intake} = P_{casing} + \rho_o \cdot g \cdot h_{above}$ (casing-head pressure plus the oil column in the annulus above the pump), with $P_{wf} \approx P_{intake}$. Pump discharge $P_{discharge} = P_{wh} + \rho_o \cdot g \cdot D_p + \Delta P_f$ (tubing side; friction adds when flow is upward). v1 does not compute these; it holds drawdown constant (open item 3).

Limitation: linear productivity index assumed; non-linear Vogel-type inflow at high drawdown is out of scope for v1.

5 · VISCOSITY → PUMP SPEED LIMIT [ASSUMPTION — STATIC FORCE BALANCE, OURS, NEEDS CALIBRATION]

$$N_{\max} = \min(30 \cdot v_f / S, 12 \text{ SPM}) \quad \text{where} \quad v_f = (\rho_s - \rho_o) \cdot g \cdot d_r^2 \cdot \ln(d_t / d_r) / (8\mu)$$

Rod weight against viscous drag gives the fall speed available; the pump allows a half-stroke period of 30/N seconds to complete it. If required exceeds available, the string goes slack. μ is in Pa·s here (1 cP = 0.001 Pa·s). With the table values, $N_{\max} = 128,240 / (\mu_{\text{cP}} \cdot S)$. The 12 SPM mechanical ceiling is tier D (the top of the N range).

Worked ($S = 2.5 \text{ m}$): day 0–60 → no binding limit · day 90 → **11.5 SPM** · fully cold → **4.5 SPM**

Key finding for the deck: the limit does not bite for about the first three months of a cycle. In the worked case it first falls below 12 SPM on day 88 (60.4 °C, 4,275 cP); across strokes of 1–3 m that day ranges from 83 to 169. Then it collapses rapidly. That end-of-cycle window is where the mechanical damage occurs, and it is precisely the period nobody currently monitors.

Floating risk [assumption, tier D]: risk = $N / N_{\max}$. Above 1, the set speed exceeds the safe speed and the rods float.

6 · ROD DYNAMICS AND CARD DIAGNOSTICS [PHYSICS — LUMPED, V1]

$$M_r \cdot \ddot{x} + C_r \cdot \dot{x} + K_r \cdot x = F_{\text{ext}}(t)$$

M_r , C_r and K_r are the rod string's effective mass, damping and stiffness. Generates a predicted dynamometer card. Comparing predicted against observed gives a residual $R_{\text{card}} = D_{\text{observed}} - D_{\text{predicted}}$ used to classify underfill, valve faults, leakage and mechanical anomaly.

Note: equation 5 answers the floating question analytically without integrating this, so floating detection need not depend on card fidelity. Reserve this model for fault classification, where card *shape* genuinely carries the signal.

7 · CSS PHASES — INJECTION AND SOAK [ASSUMPTION, TIER D]

Production (equation 2) starts from a fixed state: the heated zone at T_s with volume V_h . Injection and soak must end in that state. They are modelled to be plausible, not precise.

Injection length: $t_{\text{inj}} = 1,000 \cdot V_s / (\dot{m}_s \cdot 24)$ days with $\dot{m}_s = 3,100 \text{ kg/h}$ (OIL, tier B). 2,000 m³ takes 26.9 days; the field practice of 14–21 days corresponds to 1,040–1,560 m³.

Injection temperature: $T(t) = T_i + (T_s - T_i)(1 - e^{-t/\tau_{\text{inj}}})$, $\tau_{\text{inj}} = 1 \text{ day}$.

Heated volume during injection grows in proportion to injected heat: $V(t) = f_{\text{inj}} \cdot V_h \cdot t / t_{\text{inj}}$, with $f_{\text{inj}} = [c_p(T_s - T_i) + f_c \cdot x \cdot L] / [x \cdot L + c_p(T_s - T_i)]$ and $f_c = 0.5$. This is the sensible heat plus the latent heat of the half of the steam that has condensed by the end of injection. $f_{\text{inj}} = 0.705$ in the worked case.

Soak: T holds at T_s while the remaining steam condenses at saturation temperature, and V grows linearly from $f_{\text{inj}} \cdot V_h$ to V_h by the end of soak.

Production length: 120 days by default (t_{cut} range 60–365 days). **Days since steaming** = days since injection stopped (soak days plus production days).

v1 does not model the soak optimum. Soak time moves the calendar but does not change the production-start state.

8 · MEASURED SIGNALS [ASSUMPTION, TIER D]

A later stage infers the hidden state from signals a sensor can read. The forward model must therefore produce those signals.

Measured: wellhead temperature, pump load, flow rate q , set SPM and stroke, days since steaming. **Inferred:** reservoir temperature, viscosity, heated-zone size, floating risk.

Wellhead temperature (Ramey, 1962):

$$T_{wh} = T_{g,0} + g_G \cdot A - (T_{g,0} + g_G \cdot D + g_G \cdot A - T_r) \cdot e^{-D/A}$$

$g_G = (T_i - T_{g,0})/D$, surface temperature $T_{g,0} = 30$ °C, $D = 1,150$ m, relaxation length $A = a \cdot q$ with $a = 20$ m per bbl/d. Oil leaves the reservoir at T_r and cools towards the geotherm as it rises; a low rate cools more. Defined only while producing; no value during injection and soak.

Pump load — peak polished-rod load:

$$PPRL = W_{rb} \cdot (1 + S \cdot N^2/1790) + \rho_o \cdot g \cdot D_p \cdot A_p + 2\pi \cdot \mu \cdot v \cdot D_p / \ln(d_t/d_r)$$

$W_{rb} = (\rho_s - \rho_o) \cdot g \cdot (\pi d_r^2/4) \cdot D_p$ is the buoyant rod weight (28.8 kN). The second term is the fluid load on the plunger for a pumped-off well (16.1 kN). The third is viscous drag on the rods during the upstroke at the mean speed $v = S \cdot N/30$ (μ in Pa·s), using the same Couette drag as equation 5. $S \cdot N^2/1790$ is the Mills acceleration factor with S in metres.

Heated-zone size stays at V_h through production; only its temperature falls.

WORKED PRODUCTION CASE — ALL EQUATIONS

$V_s = 2,000$ m³, $x = 0.70$, $T_s = 250$ °C, $S = 2.5$ m, $N = 6$ SPM, all other values from the table in section five. Days are production days. Use this table to verify an implementation.

Day	T (°C)	μ (cP)	N_{max} (SPM)	q (bbl/d)	T_{wh} (°C)	PPRL (kN)	Risk
0	250.0	4.8	12.00	32.3	73.0	46.4	0.50
30	123.6	93.1	12.00	31.7	51.3	46.7	0.50
60	77.1	1,148	12.00	26.4	41.2	50.2	0.50
90	60.0	4,465	11.49	17.3	36.2	61.4	0.52
120	53.7	7,998	6.41	12.7	34.4	73.2	0.94

Heated zone 4,338 m³, $r_h = 11.8$ m, $\tau = 30$ d, injection 26.9 days. $N_p = 2,946$ bbl, SOR = 4.27. At $N = 6$ SPM the rods start to float on production day 127; at $N = 8$ SPM, on day 106.

Where ML attaches: Corrected = Physics + ML_residual, applied at the wellbore and dynamometer modules. Physics stays the backbone; models learn the discrepancy rather than replacing the mechanism. Two models fall out — one predicting the temperature decay curve for this well and cycle, one solving the constrained optimisation in section three.

Five Parameters for model training

Confidence tiers — **A** stated in the problem statement, **B** published Oil India or SPE data, **D** our assumption, requires calibration. **std** marks physical constants and steam-table values.

Symbol	Meaning	Unit	Value / range	Conf.
$^{\circ}\text{API}$	Crude gravity	$^{\circ}\text{API}$	17–19 (PS) vs 14–17 (field); model uses SG 0.963 $\approx$ 15.4	A / B
T_i	Native reservoir temperature	$^{\circ}\text{C}$	46–48 (PS) vs $\sim$ 50 BHT (field); model uses 50	A / B
μ_o	Viscosity at 50 $^{\circ}\text{C}$, at 50 s^{-1}	cP	10,000–13,000; model anchor 11,500 (midpoint)	B
D	Well depth, Jodhpur Sandstone	m	$\sim$ 1150	B
P_r	Reservoir pressure	ksc / MPa	$\sim$ 116 ksc = 11.4 MPa UNCERTAIN — CONFIRM	B
ϕ	Porosity	fraction	< 0.10 CONFIRM SOURCE — unusually low for this sandstone; no equation uses it	B
V_s	Steam volume per cycle	m^3 CWE	Literature 300–12,000; Baghewala field 1,040–1,560 (3,100 kg/h for 14–21 days); simulator 500–4,000, default 2,000	B
$\dot{m}_s$	Steam injection rate	kg/h	$\sim$ 3,100	B
t_{inj}	Injection period	days	14–21 (field); model: $V_s/\dot{m}_s$	B
x	Steam quality at wellhead	fraction	0.60–0.80 (literature); 0.60–0.70 (field)	B
T_s	Steam temperature	$^{\circ}\text{C}$	200–300 (literature); 280–305 surface (field); model 250 effective downhole	B / D
P_{inj}	Injection pressure	MPa	1–11 ($\leq$ 1600 psig preferred); field steam 85–97 ksc (8.3–9.5 MPa)	B
t_{soak}	Soak time	days	3–14 (field: 50–60% of injection time)	B
t_{cut}	Production length before re-steam	days	60–365; model default 120	B / D
N	Pump speed	SPM	2–12; 12 is the mechanical ceiling	D
S	Stroke length	m	1.0–3.0	D
SOR	Steam–oil ratio	—	3.0–5.2	B
RF	Recovery factor	%	< 20 typical, up to 35	B
q_o	Field production rate (whole field)	bbl/day	$\sim$ 655 (July 2025)	B
ρ_w, c_p, L	Water density, specific heat, latent heat at 250 $^{\circ}\text{C}$	kg/m^3 , $\text{kJ}/(\text{kg}\cdot\text{K})$, kJ/kg	$1,000 \cdot 4.187 \cdot 1,716.6$	std
M_R	Rock volumetric heat capacity	$\text{J}/(\text{m}^3\cdot\text{K})$	2.35×10^6	D

Symbol	Meaning	Unit	Value / range	Conf.
η	Thermal efficiency	fraction	0.4–0.7; model 0.5	D
h	Net heated thickness	m	9.92 (back-solved from $r_h = 11.8$ m)	D
τ	Cooling time constant	days	$\tau_\infty \cdot r_h / (r_h + h)$, $\tau_\infty = 55.2$; 21–35 over the simulator range	D
τ_{inj}, f_c	Injection heating constant; steam condensed by end of injection	days, fraction	1 · 0.5	D
A, B	Walther constants	—	$A = 9.3933$ (computed from the anchor), $B = 3.5$	D
ρ_o / ρ_s	Oil / steel density	kg/m ³	963 / 7850 (SG 0.963)	D
d_r / d_t	Rod / tubing ID	m	0.0222 (7/8") / 0.062 (2 7/8")	D
r_w / r_e	Well radius / drainage radius	m	0.1 / 100	D
q_{cold}	Unstimulated (cold) rate per well	bbl/day	10	D
A_p / η_v	Plunger area (1.75") / volumetric efficiency	m ² / fraction	1.552×10^{-3} / 0.8	D
D_p	Pump setting depth	m	1,100	D
$T_{g,0} / a$	Surface temperature / Ramey relaxation constant	°C / m per bbl/d	30 / 20	D

Conversions — 1 ksc = 98.07 kPa · 1 bbl = 0.159 m³ · 1 psi = 6.895 kPa · 1 cP = 0.001 Pa·s · $v(cSt) = \mu(cP)/SG$ · $SG = 141.5/(131.5 + ^\circ API)$ · the model uses $SG = 0.963$ throughout.

State vector for the twin: $X = [T_r, V_h, P_r, \mu_o, \lambda_o, q_{in}]$ with control input $u = [V_s, x, t_{soak}, N, S]$ and uncertainty σ . Every parameter carries value, unit, source, calibration method and confidence.

On the conflicting values

API gravity and native temperature differ between the problem statement and published field data. Preserve both rather than silently reconciling — but note that the field figures come from Oil India's own technical publications and SPE papers written by the field's operators, while the problem statement is an outreach document. Model with the field values; cite the discrepancy openly if asked. The model uses $T_i = 50$ °C and $SG = 0.963$.

Six

Open items

1 · Second viscosity measurement

One reported value exists: 10,000–13,000 cP at 50 °C, and the model uses the 11,500 cP midpoint. Fitting Walther requires two points, so $B = 3.5$ is assumed. Because the curve is anchored at 50 °C, every B gives the same viscosity there. The spread between $B = 3.2$ and $B = 3.8$ (the higher viscosity divided by the lower, minus one) is 0% at the anchor. It grows in both directions — 57% at 30 °C ($B = 3.2$ gives 96,700 cP, $B = 3.8$ gives 151,700 cP) and 78% at 250 °C. So the uncertainty is smallest near reservoir temperature, where the well spends most of its life, and largest at steam temperature.

Inside the cycle, measured the same way, the spread is 41% at 77 °C and 77% at 124 °C. B from 3.2 to 3.8 moves the day the float limit binds only from day 86 to day 91, and changes the worked SOR from 4.35 to 4.20. A second measurement, ideally near 100 °C, would fix B and test whether the Walther form holds for this crude.

2 · Calibrate the floating limit against failure history

Equation 5 is a static force balance. It ignores rod stretch, elastic wave effects, plunger fluid load and mechanical friction, so it reads optimistic — real $N_{\max}$ is lower. Two more reasons it reads optimistic: it uses the heated-zone viscosity for the oil around the rods, but oil cools as it rises up the tubing, so the upper rods sit in thicker oil; and it compares the rod's fall speed with the mean downstroke speed $S \cdot N/30$, while for near-sinusoidal motion the peak speed is $\pi/2$ higher — on that basis the day-90 limit would be 7.3 SPM, not 11.5.

Oil India states it holds rod failure and pump unsettling history. Fit one correction factor so predicted $N_{\max}$ aligns with speeds at which failures actually occurred, and present it as calibrated rather than first-principles.

3 · Reservoir pressure has no governing dynamic

P_r appears in every state vector and drives q_{in} directly, but no dP_r/dt relationship is defined. Either add a simple depletion and support term, or state explicitly that it is held constant or externally supplied for $v1$, with the reason. $v1$ answer: P_r is held constant, and the rate law in equation 4 holds drawdown constant, so P_r does not enter any worked number.

The reported value is also uncertain. 116 ksc at 1,150 m is close to hydrostatic (about 115 ksc), which does not fit the problem statement's "low reservoir pressure". Surface steam at 85–97 ksc could not enter a reservoir at 116 ksc. The figure may be the initial pressure. Confirm it with Oil India before using it; it is not corrected here.

Unmodelled, worth naming to judges: the crude is non-Newtonian — viscosity is quoted at a shear rate of 50 s^{-1} , so thickness depends on shear as well as temperature. Since rod motion through the fluid *is* a shear event, pump speed influences viscosity, which influences the safe pump speed. A feedback loop inside equation 5. Acknowledge as a known refinement; do not build it until the main chain works.